1 Bit Is All You Need

The Weakest Superpower: No Unaccounted Differences

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You may hide a difference. You may not leave it without a provenance.

Abstract

I made a bet with God: make the world as hard as you like, just don't skip out on the bill. The one bit of the title is a binary question: can an actual difference acquire actual standing without accounting for any structural provenance? We prove the answer is no, and make it pay out three times. First, between any two lawful world states of any source-native authoritative root, and any two states of any complete living process, every real difference acquires a positive structural identity from distinct registered occurrences, so external supplements and magic bits are empty—no-magic is a derived theorem, not a preset field. Second, the completion of one fixed mother source's finite-programme image forms every lawful world, every complete inquiry, and every complete macro-process on the original interface and recovers all their original operations; the original physical process consumes this formation directly. Third, at every cofinal admission point the original ledger turns the complete future residual into the next request, pays the same debt strictly with its own syntax, and hands over an answer, a receipt, and the next ledger; when language runs out, failure generates a minimal new direction, and old debt and new responsibility are each disposed of in the same first write. The paper ends by handing every conclusion to an equality the reader can check: if you want to break it, break it there.

Keywords: no unaccounted difference; total reality; structural identity; registered occurrence; fixed-mother formation; source-native payment; minimal revision; dependent types

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1 I Have One Very Weak Superpower

I made a bet with God: make the world as hard as you like, just don't skip out on the bill.

My stake is a single superpower, and the weakest one on record. Faced with any actual difference, I may ask one question: **where did you come from?** I cannot foresee tomorrow, rewrite yesterday, or choose anyone's answer for them. I refuse exactly one reply: "It has no origin; it just is."

That refusal has two values, so it is one bit. What can something this weak possibly do?

First, look at what it stops. Suppose a source root has registered a lawful world state w. Anyone may tape a Boolean next to it and obtain (w, false) and (w, true). As raw data the two really differ; through the admission projection (the first component, the only one admission accepts), they are the same world state, the same registered occurrence, the same authoritative evolution (§2.2, Figure 2). Both facts are true. The overreach is the step in between: **one more raw bit, therefore one more reality.** That step needs a licence, and the Boolean brought none.

A principle that can only refuse is worth little. What this bit actually does is pay out three times, each payout actually called.

This paper's contribution. The paper organizes identity, formation, action, and revision as four interfaces of one criterion: after a change of representation, an execution, or a revision of language, the original occurrence, responsibility row, and actual successor must still connect item by item. The Boolean ghost counterexample (§2), the separation of occurred facts, observation, and future (§3), and the location of the claim a failure requires us to correct (§8) supply the full argument and checking method for this criterion.

The mathematical constructions retain their original attribution: identity comes from the zero-law and total-reality core, reused in Whole-Ledger Accounting [8]; formation from Theorem 8.4 and Corollary 8.5 of the second edition of the physics flagship [7]; payment from Theorems 9.1–9.3 of Whole-Ledger Accounting; and revision from Theorems 8.4 and 8.6–8.7 of the second edition of the process core [6] and Theorem 12.3 of Whole-Ledger Accounting. The common conclusions come first; the sections below develop the exact contracts and proofs.

Main Theorem 1.1 (Three payouts of one bit).

1. **Identity: every real difference has an occurrence identity.** For every relation network N, vocabulary V, and source-native authoritative root R, any two unequal lawful world states of R—including finite visits, cofinal landing visits, and their successors—have distinct registered occurrences. Actuality and stable reference are generated by the root's compiler receipts and occurrences; total reality then excludes external supplements and magic bits. Any two unequal states of any complete living source process likewise have distinct complete registered currents, with actuality and stable reference generated by that process; no input of NoMagic type occurs (Theorems 2.6, 2.8).
2. **Formation: one fixed mother forms three complete object domains.** Fix the physical original-occurrence root $S_\star$. For any N, V, R and any lawful world state w of that root on the original Type0 interface, any complete inquiry q, and any native inquiry-engine process P, there are a material rank and a material in the completion of the finite-programme image of $S_\star$ such that the actual factory forms the object and recovers all its original operations from that same material, including occurrence, whole ledger, successor, every query compilation,

activation law, and runtime history at every natural-number depth. Every formed world carries the total reality of item 1; addresses and ranks are generated inside the proof (Theorem 4.2).

3. **Action and revision: the original responsibility enters an actual next step.** For any original source factory $\mathcal{F}$, complete initial frame I, programme cfg, and language identity, at every natural-number cofinal admission point produced by the original run, the literal residual and paid boundary are equivalent at every finite allowed action word and every original site, and the complete material of the same next current generates the original request. At every admission point of $\widehat{\mathcal{F}}$, derived from $\mathcal{F}$ inside the source by installing the complete pair reading of that same current, the original syntax strictly pays the same debt without refill; after the original suffix completes, one action delivers the completion value, an answer with its receipt, the destinations of the old and born rows, and the canonical next (Theorems 5.1–5.3). A rooted failure with empty expression and realization fibres generates the minimal new direction; every field-grounded revision gives the old and new rows one first write, and the new root generates their separate answers and nexts (Theorems 6.2, 6.5).

Each item is paid by its own construction, and they meet on one kind of object: the registered occurrence of item 1 is a field recovered after formation in item 2, and the payment and revision of item 3 land on the registered occurrences, ledger writes, and successors of the same root. Figure 1 draws these connections.

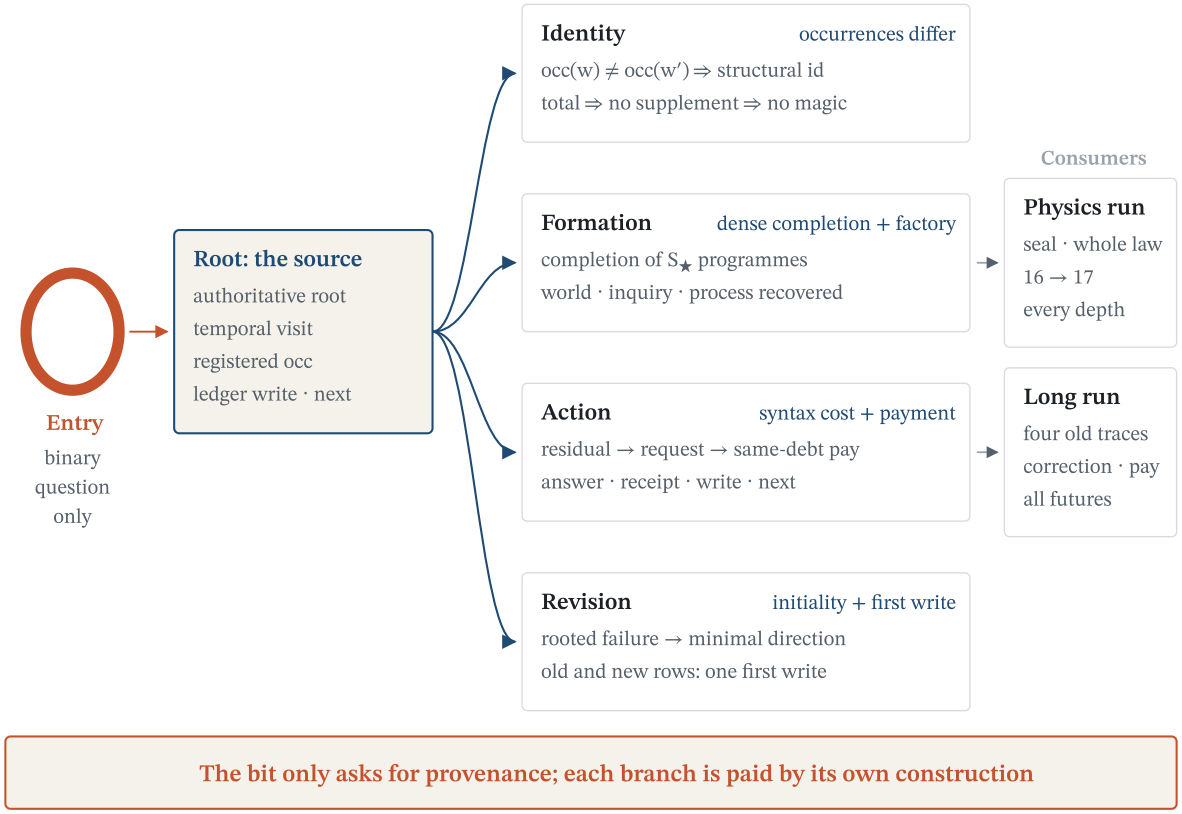


Figure 1 One 0, a whole forest. The vermilion 0 on the left is only the entry stamp: the binary question whether an unaccounted difference can acquire standing. The actual root is the source and its registered occurrence. Each of the four branches names the construction that pays for it: unequal registered occurrences give structural identity and exclude external supplements and magic bits; the completion of the fixed mother's finite-programme image forms the world, inquiry, and macro-process domains of the original interface and is consumed directly by the original physical process; the complete future residual generates a request, a same-debt payment, and a completed answer; a rooted failure generates the minimal direction, and revision sends old and new rows into one first write. The original physical process and the long-computation construction in the right column only consume the constructions of the branches they attach to and bear none of their load. Arrows denote construction or direct invocation, not temporal order.

1.1 What exactly is this bit?

The "1 bit" of the title is defined once. It is a binary question: **can an actual difference acquire actual standing without accounting for any structural provenance?** For the source structures listed here we prove the answer is "no"; that is the one bit of the paper.

This bit is the value of a question, not an amount of data: it does not encode the universe in one binary digit (a random variable that always takes the same value has zero Shannon entropy [3]). The formal MagicBitAt is a structure at the level of propositions (Definition 2.4); we prove it empty under specified semantics, starting from positive classification, and no construction takes "no magic" as a field.

Writing the question as a bit makes it a blade. Its edge lies on every equality that follows: unequal registered occurrences (§2), injective complete currents (§3), restriction recovering the original after formation (§4), residual and paid boundary equivalent at every future reading (§5), old and new rows sharing one first write (§6).

1.2 Reading route

Choose a route by what you want:

- **Just the conclusions**: read §§1–3. Main Theorem 1.1, the Boolean ghost, the derivation of total reality, and the separation "yesterday is rigid, tomorrow is not inventory" are all there, with no type-theory background needed.
- **How the constructions work**: continue with §§4–6. §4 lets one fixed mother form every object of the original interface, §5 uses a square with nonzero effect to show how the original ledger pays, answers, and turns the page, and §6 handles the case where language runs out; complete proof outlines are in Appendices A–C. §7 summarizes the wiring of a long computation; the analytic objects and full contract are in Appendix E.
- **Want to break it**: go straight to §8, then use Appendix D to hand each claim to its original declaration and executable check. §9 returns to the opening bet.

From §4 on, three words recur. **Transport**: moving an object along a type equality into the type on the other side. **Heterogeneous equality**: the two sides have different types and are equal after transport, written $\doteq$. **Dependent direct sum**: each site has its own carrier, assembled into one whole; the embedding ι_j places an element of site j into that whole.

2 This Bit, Show Me Your Papers

This section starts from admission: a lawful state is a source-generated temporal visit, and its registered occurrence retains that visit itself. Recovering the state from the occurrence therefore takes one projection; at process level, the same conclusion consumes the injective admission of complete currents. The construction cost comes before admission: the root must supply complete event inventory, actual emission, and commuting compilation before an admitted state can read authoritative evolution, whole-ledger write-back, and successor from the same visit. We spell out that structure first, then derive no magic from it.

2.1 Lawful world states are registered visits

First fix the domain of discussion. A **source-native authoritative root** R is built on a relation network N and a vocabulary V: it consists of an authority source (restructuring ledger source, complete event-inventory admission, law surface, and projection law), an emitter issuing the original event at every current, and a proof that compilation commutes with the emitter. Complete event-inventory admission requires a bidirectional constructive correspondence between the presentation and the concrete source's currents and complete event families at each current, with both composites identities [8]. There is no field about magic here.

Definition 2.1 (Temporal reachability and lawful world states). A **temporal reachability witness** of a current c has exactly two constructors:

- **finite**: starting from the root's initial visit, each step advances along the successor equation $\mathsf{next}(c_i) = c_{i+1}$ given by the original compiler until c is reached;

- **post-cofinal**: starting from a source-generated cofinal landing visit, finitely many further steps along the same successor equations reach c.

A **lawful world state** of R is a pair $w = (c, h)$ with h a temporal reachability witness of c; the collection is $\mathsf{Law}(R)$. The **registered occurrence** of w is the dependent pair

$$\mathsf{occ}(w) = \langle w,\ \varepsilon_R(w) \rangle \in \sum_{u:\mathsf{Law}(R)} E_R(u), \tag{1}$$

where $\varepsilon_R(w)$ is the exact temporal causal event generated by the root at w. The same visit also reads off the authoritative temporal evolution $\mathsf{auth}(w)$ and a positive disposition from the same compiler image: one of native write, relation write, continued transport, redirect, or local faithful terminal—one of five.

The admission surface of the zero law is this carrier itself: no second registry, no caller-submitted world record. A post-cofinal history can begin only at a source-generated cofinal landing, so a finite visit cannot pose as a cofinal one; the finite embedding is injective, so the same current reached along different causal histories remains two lawful states.

This admission retains operational capacity: the two-way complete-inventory correspondence recovers the original currents and the event family at each current; the root's emitter and commuting compilation then supply the actual evolution at the visit. §4 recovers these fields from one mother's material, §5 makes the original row pay and generate a successor within one whole-ledger write, and §6 carries the old responsibility across revision. Occurrence identity is a field these operations carry together.

Lemma 2.2 (World identity is occurrence identity). For $w, w' \in \mathsf{Law}(R)$, if $\mathsf{occ}(w) = \mathsf{occ}(w')$ then $w = w'$. Equivalently, $w \neq w'$ implies $\mathsf{occ}(w) \neq \mathsf{occ}(w')$.

Proof. Take the first projection of the dependent pair (1). The contrapositive is the same fact. □

The proof is one line because admission has already paid the whole cost: the type of lawful states contains only registered visits, and there is no unregistered component in which anything could hide.

2.2 Two Booleans, one world

Now let the Boolean ghost enter. Take the concrete inquiry world in the fixed research source Y and write its root as $R_\circ$; let v_1 be the initial visit, v_2 the visit obtained from v_1 by one original successor step, and $w_i = \mathrm{finite}(v_i)$.

Proposition 2.3 (One more raw bit, no more reality).

1. **A real difference has an identity.** $v_1 \neq v_2$, hence $w_1 \neq w_2$ and $\mathsf{occ}(w_1) \neq \mathsf{occ}(w_2)$.
2. **The ghost bit has none.** On $G = \mathsf{Law}(R_\circ) \times \mathrm{Bool}$ let $g_0 = (w_1, \mathrm{false})$ and $g_1 = (w_1, \mathrm{true})$. Then $g_0 \neq g_1$; yet their admission projection (first component) satisfies

$$g_0.1 = g_1.1, \qquad \mathsf{occ}(g_0.1) = \mathsf{occ}(g_1.1), \qquad \mathsf{auth}(g_0.1) = \mathsf{auth}(g_1.1). \tag{2}$$

3. **The producer has no ghost argument.** The two constructors of lawful states accept only a finite root visit or a cofinal visit history; there is no lawful-state constructor accepting a visit together with a Boolean.
4. **An unadmitted raw root does not get in.** A mixed terminal/write root that has only a raw ledger compiler, without complete event-inventory admission, cannot occur as the root of a lawful world state.

Proof. (1) The histories of the two visits are the initial constructor and the one-step constructor, which differ, so the visits differ; injectivity of the finite embedding gives $w_1 \neq w_2$, and Lemma 2.2 gives distinct occurrences. (2) If $g_0 = g_1$, the second projection gives false = true, a contradiction; the three equations in (2) hold by definition, since both first components are the same w_1. (3)(4) are facts at the level of types: the former follows from the constructor signatures in Definition 2.1, the latter from the complete inventory proof the admission type demands; the original formal control checks that both overreaching terms fail to type-check. □

There are two true facts here, and neither may be dropped. G is a lawful mathematical object, and $g_0 \neq g_1$ is true; after admission there is only one world, and that is true too. What is wrong is promoting the former to the latter. For the Boolean to become a real difference, it must first change the source or root occurrence, or be registered by a later residual or revision event—and then, as in (1), it enters with a different registered occurrence. If it is forever indistinguishable, untransportable, and effectless in every source-fixed operational context, it is only another presentation of the same reality.

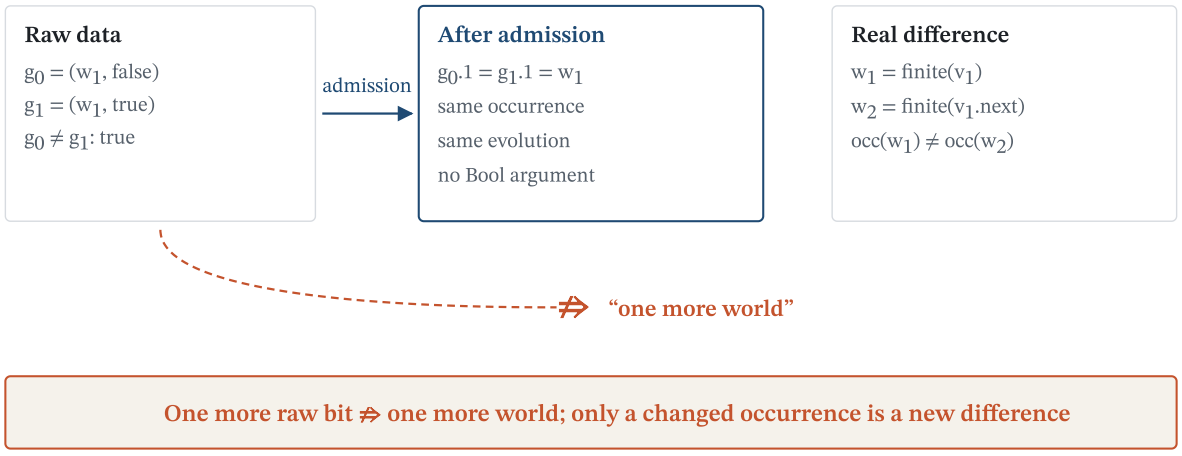


Figure 2 Two bits, one identity. Left: as pairs, the raw data (w_1, false) and (w_1, true) really are unequal. Middle: the admission projection (first component) sends both to the same lawful state w_1, with registered occurrence and authoritative evolution equal item by item; the constructors of lawful states have no place for a Boolean. Right: the real difference $w_1 \neq w_2$ comes from two distinct root visits, whose registered occurrences differ. Vermilion marks the overreach: inequality of raw data cannot be promoted directly to inequality of worlds.

2.3 Deriving no magic in a vocabulary that never mentions magic

The derivation below does not start from "no magic". It first gives five weak notions that can be realized independently, then proves that a magic claim generates an external-supplement obligation

by itself, and finally excludes that obligation by positive classification.

Definition 2.4 (Difference semantics, internal determination, external supplement, and magic bit). A **difference semantics** $\mathcal{R}$ consists of a type D with five predicates: actuality Act, stable reference Ref, real difference Real, redundant presentation Red, and structural identity Id. For $d \in D$:

- **actual difference** $\mathsf{AD}(d)$: $\mathsf{Act}(d) \wedge \mathsf{Ref}(d) \wedge \mathsf{Real}(d)$;
- **internal determination** $\mathsf{ID}(d)$: $\mathsf{AD}(d) \wedge \mathsf{Id}(d)$, an actual difference together with its positive structural identity;
- **external supplement** $\mathsf{ES}(d)$: $\mathsf{AD}(d) \wedge \neg\mathsf{Red}(d) \wedge \neg\mathsf{ID}(d)$;
- **magic bit** $\mathsf{MB}(d)$: the same three fields—actual, nonredundant, yet refusing every internal determination.

The **total reality** $\mathsf{Total}(\mathcal{R})$ of $\mathcal{R}$ is the positive classification

$$\forall d \in D, \quad \mathsf{AD}(d) \implies \mathsf{Red}(d) \vee \mathsf{Id}(d). \tag{3}$$

ID is not an alias for "not magic"; it carries the structural identity itself. And (3) only says where each actual difference falls, with no negative field. That magic bits and external supplements have the same fields is not a duplicated definition: the former is a claim, the latter the obligation that claim immediately incurs—**to claim that it is real and unsourced is to claim it must be written in from outside** ($\mathsf{MB}(d) \to \mathsf{ES}(d)$ field by field).

Theorem 2.5 (Total reality excludes external supplements and magic bits). If $\mathsf{Total}(\mathcal{R})$, then for every $d \in D$ both $\mathsf{ES}(d)$ and $\mathsf{MB}(d)$ are empty.

Proof. Suppose $\mathsf{ES}(d)$. Hand its actual-difference component to (3). If it lands in $\mathsf{Red}(d)$, this contradicts nonredundancy; if in $\mathsf{Id}(d)$, together with the actual difference it forms $\mathsf{ID}(d)$, contradicting "not internally determined". A magic bit first becomes an external supplement and then meets the previous step. The proof is constructive; both branches use only the witnesses given, with no excluded middle, quotient equality, or choice. □

Theorem 2.5 by itself grants no world authority: if a caller simply hands over $\mathsf{Total}(\mathcal{R})$, it is merely a parameter of the theorem. Next, the root generates it itself.

2.4 The root supplies total reality itself

Theorem 2.6 (Total reality of a source-native authoritative root). For any N, V and any source-native authoritative root R, define the root semantics $\mathcal{R}_R$:

$$\begin{aligned} D &= \mathsf{Law}(R) \times \mathsf{Law}(R), \\ \mathsf{Act}(w, w') &: \text{ both sides have canonical receipts of the root's temporal-visit compiler,} \\ \mathsf{Ref}(w, w') &: \text{ both registered occurrences } \mathsf{occ}(w), \mathsf{occ}(w') \text{ exist,} \\ \mathsf{Real}(w, w') &: \ w \neq w', \qquad \mathsf{Red}(w, w') : \ \mathsf{occ}(w) = \mathsf{occ}(w'), \qquad \mathsf{Id}(w, w') : \ \mathsf{occ}(w) \neq \mathsf{occ}(w'). \end{aligned} \tag{4}$$

Then: (a) every pair of unequal lawful states is an actual difference whose actuality and stable reference are generated by the root itself; (b) $\mathsf{Total}(\mathcal{R}_R)$; (c) for every d, $\mathsf{ES}(d)$ and $\mathsf{MB}(d)$ are empty.

Proof. (a) For every lawful state the root proves that its generated temporal evolution is the canonical compiler image; one such proof for each side gives actuality; stable reference takes the two registered occurrences themselves; the real difference is the given inequality. (b) For an actual difference (w, w'), its real-difference component $w \neq w'$ gives $\mathsf{occ}(w) \neq \mathsf{occ}(w')$ by Lemma 2.2, the right disjunct of (3). (c) By Theorem 2.5. □

Three remarks give the weight of this theorem. First, the quantifiers are universal: every relation network, every vocabulary, every source-native authoritative root, and all three kinds of visit—finite, cofinal, post-cofinal. Second, the hypothesis is not vacuous: Proposition 2.3(1) already exhibits a pair of actually unequal lawful states, so (b) does not hold on an empty set. Third, there is no circle: Id is generated from admission structure by Lemma 2.2, not rewritten from "not magic". In the root semantics structural identity is exactly the negation of redundancy, so the whole content of (b) lies in Lemma 2.2, that is, in the admission carrier itself—which is why this section puts the weight on admission.

2.5 A living process crosses a local terminal, and identity goes with it

Fixing one root is not enough: a living process can hand control to another root at a local terminal, changing both vocabulary and root. At process level, identity is carried by the complete registered current.

Definition 2.7 (Complete living source process). A **living current** over a network N is a dependent record (V, ρ, v): a vocabulary V, a living root ρ with its terminal-handoff law, and a temporal visit v of ρ. A **complete living source process** $\mathcal{P} = (S, \iota, s_0, \sigma)$ consists of an arbitrary state type S, an **injective** $\iota : S \hookrightarrow \mathrm{Cur}(N)$, an initial state s_0, and a one-tick successor σ; σ comes with a proof that $\iota(\sigma s)$, after erasing the handoff law, equals the next current generated by the root of $\iota(s)$ at that visit, and that on non-terminal branches the living root itself is preserved (heterogeneously equal). At a local faithful terminal, the next current is generated by the source-fixed terminal-handoff law. [6]

Theorem 2.8 (Total reality of a complete living process). For any N and any complete living source process $\mathcal{P}$, let the process semantics have differences $S \times S$, actuality the existence of canonical causal "answer-and-next" at both sides, stable reference the existence of process-indexed causal occurrences at both sides, real difference $s \neq s'$, redundancy $\iota s = \iota s'$, and structural identity $\iota s \neq \iota s'$. Then total reality holds, and external supplements and magic bits are empty.

Proof. Actuality and stable reference are given by the process's own canonical answer-and-next and emitted events. If $s \neq s'$ and $\iota s = \iota s'$, injectivity gives $s = s'$, a contradiction; so the difference falls into structural identity. The rest is Theorem 2.5. □

Example 2.9 (A terminal is not a global stop bit). The fixed research source Y contains a concrete living process: the state type is $\mathrm{Option}\,\mathbb{N}$, $\iota(\mathrm{none})$ is the terminal visit of a local terminal root, and $\iota(\mathrm{some}\,n)$ is the visit of depth n of a native-write root (the first visit is $n = 0$); the successor is $\mathrm{none} \mapsto \mathrm{some}\,0$, $\mathrm{some}\,n \mapsto \mathrm{some}\,(n+1)$. The two vocabularies differ (the terminal

vocabulary has a faithful terminal fibre, the write vocabulary has none), and the finite depth of the write visit is exactly n, whence ι is injective. The states none and some 0 lie across the local terminal, and their difference acquires structural identity from distinct complete registered currents; the first successor, after erasure, equals exactly the handoff occurrence generated by that terminal. What separates them is not a bit saying whether the world has stopped, but two complete living currents.

2.6 Verdict of this section

"No magic" has been written into no definition here. What has been written into definitions is admission: which objects qualify as lawful world states, which qualify as the registration of process states. Total reality is then generated by admission structure, and excluding magic is its corollary.

This puts the burden of proof where it belongs. Whoever insists that some extra bit belongs to reality now has two roads: let it change some source or root occurrence, so that it enters with a new registered occurrence and acquires structural identity; or admit that it changes no registration, so that it is a presentation of the same reality. The third road—"really different, but never on the books"—is an empty type under this section's admission.

3 Yesterday Cannot Recast Its Actors; Tomorrow Cannot Steal the Script

Once an occurred fact is registered, a reader may read rigidity too far: since one complete current has only one successor, any local observation might seem to recover the whole past, or even the whole future might seem already written. This section separates three things.

3.1 One complete current, one answer and next

Theorem 3.1 (Current reflection and on-demand history). Fix any complete living source process $\mathcal{P} = (S, \iota, s_0, \sigma)$.

1. If $\iota s = \iota s'$, the canonical causal "answer-and-next" at s and at s' are heterogeneously equal; in particular $\iota(\sigma s) = \iota(\sigma s')$.
2. If $\iota(\sigma s) \neq \iota(\sigma s')$, then $\iota s \neq \iota s'$: every difference in the successor is already reflected in the complete registration of the current.
3. After erasure, $\iota(\sigma s)$ equals the next current generated by the root of $\iota(s)$, and the two roots have the same law surface.
4. For every $n \in \mathbb{N}$ and every state s, the original successor recursively generates a finite history $\mathsf{hist}(n, s)$ of length n: $\mathsf{hist}(0, s) = \mathsf{exhausted}(s)$, $\mathsf{hist}(n+1, s) = \mathsf{step}(\mathsf{hist}(n, \sigma s))$. The history type has only these two constructors.

Proof. (1) Injectivity gives $s = s'$; substitute. (2) is the contrapositive of (1). (3) The first half is the first component of the successor proof; for the second, apply the law surface to it and use the root property that generating the next current preserves the law surface. (4) Recursion on n. □

(1) and (2) close one particular back door: **first assume, in a theorem premise, a fact that decides the outcome, then repackage "provable under that assumption" as "that fact has occurred in this world".** In a fixed process, an external premise that would change the canonical answer or successor must change the registered complete current or set up another source process; it cannot quietly alter tomorrow under the same complete current. Conditional reasoning stays free; what it produces is a conditional proposition, not the next step of this world.

(2) says that a fixed-law process cannot hide a theory revision inside a local terminal or a carrier handoff; revision has its own independent construction (§6). In (4), exhausted only means the observation budget is used up, not that the world ended: the history type has no "stop" constructor. Taking $n = 2$ from none in the process of Example 2.9: the first row is the local terminal's answer, the second already sits on the first visit some 0 of the write root, and the history reaches some 1.

3.2 Unseen is not absent

Rigidity belongs to the complete current. An observation is a map from complete states to less information, and it may read different states as one value.

Definition 3.2 (Safety relative to a consumer, hidden difference). Let $q : A \to O$ be an observation and $c : A \to K$ a consumer. Call q **safe** for c if $\forall x, y,\ q(x) = q(y) \Rightarrow c(x) = c(y)$. A **hidden-difference witness** for q relative to c is a pair x, y together with $q(x) = q(y)$ and $c(x) \neq c(y)$.

Theorem 3.3 (A hidden difference cannot be certified safe). Given a hidden-difference witness, "q is safe for c" is empty; every further compression $h \circ q$ is refuted by the same pair. In a native responsibility process, if the scope of an actual commitment admission is invisible to some scope observer, that observer reads the same list before and after admission while the complete current-responsibility observer reads lists of different lengths; the admission event, the invisibility equation, and this complete difference together generate a full-trace redirect that keeps the original event.

Proof. For the first half, substituting the witness into the safety definition gives a contradiction; compression only needs h applied to the observation equation. For the second: the scope observer filters out live responsibilities of invisible scope, so the newly admitted one is filtered and the list is unchanged; the complete observer reads every slot as a live responsibility or an empty slot, and admission adds a slot, so the lengths differ. The redirect record packages this actual admission event directly with its hidden witness. □

The consumer in this theorem is concrete: the complete current-responsibility observer. "The interface says nothing changed" thus has three distinct statuses: actually safe for the present consumer; refuted by a hidden difference; not yet classified. The third is not a polite spelling of the first.

The same criterion has a stronger instance in arithmetic. An original arithmetic chain of Whole-Ledger Accounting constructs a nonzero class z in a globally compatible integer cokernel G: with F the family of all finite-stage complexified observations and J the original stage-one integer

consumer, $F(z) = F(0)$ while $J(z) \neq J(0)$; so there is no function h with $J = h \circ F$, nor any factorization through the observable quotient [8] (Theorem 13.3). What fails to see this one nonzero class is the whole family of finite-stage complex readings, not merely some coarse reading. Appendix A.4 writes out the structure of the argument.

3.3 Tomorrow is not inventory; it is responsibility

That the complete current decides the next step does not mean that every concrete fact after it already exists as data. Theorem 3.1(4) supplies the ability to generate prefixes on demand for every finite n; the process stores no completed execution table. The complete future model of §5 is likewise: it gives an equivalence criterion over all finite action words and all sites, the criterion itself universal, while concrete future events are still generated by the actual run.

This matches a plain distinction: what present reality must own is the current state together with its law of generated succession, (s_t, σ), not $(s_t, s_{t+1}, s_{t+2}, \dots)$. If all future facts were laid out as independent data, whatever part of them was not generated by the process would be preloaded magic information. **This bit does not foretell tomorrow for me. It makes sure tomorrow is not free.**

4 One Mother Forms All Three Object Domains

The previous two sections hold for every given root and process. The reader may press: where do these roots come from? If every world, every inquiry, every process required a separately chosen source, "which one" would itself be an unaccounted difference. The answer of this section: no separate choice is needed.

The construction has three steps. **Finite approximation**: for a finite input set, a finite output prefix, and a precision, one finite-programme visit of the same mother meets all these tests. **Dense completion**: complete the finite-programme image in the pointwise uniform structure, obtaining a reading onto the full function space. **Factory recovery**: combine all the original object's materials in one witness and let the actual factory form the object from that material and recover its operations item by item. Together these steps give the meaning of forming every complete object below.

4.1 Finite programmes and their completion

Fix the physical original-occurrence root $S_\star$, the authoritative ledger root of the SpinPair living root of the physics flagship [7]. Let $\mathbb{S} = \mathbb{R}^{\mathbb{N}}$ be real streams; at material rank r, let X_r be the input carrier and $\mathcal{A}_r$ the observation addresses separating inputs. A finite programme consists of a finite address table and one lawful visit of $S_\star$; evaluation pads observed values to a stream and passes it to the original mother finite evaluator:

$$\begin{aligned} \mathsf{Prog}_r &= \coprod_{n\in\mathbb{N}} \mathcal{A}_r^n \times \mathsf{Visit}(S_\star), \\ L_r((a_0,\dots,a_{n-1};\nu),x) &= \mathsf{finiteLaw}_{S_\star}(\nu,\ \mathrm{pad}_n(\mathrm{obs}_r(a_i,x))_{i<n}), \\ \mathcal{R}_r &= \operatorname{im} L_r, \qquad \mathcal{M}_r = \widehat{\mathcal{R}_r}. \end{aligned} \tag{5}$$

The completion uses the uniform structure input by input and finite output coordinate by coordinate. Low-rank and high-rank materials are both formed by (5); high-rank materials may take already formed complete materials as inputs.

The approximation quantifiers here are: for every target function, every finite input set, every finite output prefix, and every positive precision, there is a finite programme meeting that test. Completion supplies the complete material, and the bijective completion reading recovers the complete function. Appendix B.1 proves this passage; B.2–B.3 then connect it to the factories for complete worlds, inquiries, and processes.

Definition 4.1 (Formation). A material $m \in \mathcal{M}_r$ is **formed by** $S_\star$ if m lies in the closure in $\mathcal{M}_r$ of the finite-programme image $\{\iota_r(L_r(p)) : p \in \mathsf{Prog}_r\}$.

Formation is a conclusion, not an admission condition: the finite-programme image equals the raw image, which is dense in its completion, so **every** material satisfies Definition 4.1. The real work comes after: constructing a material for each original object and letting the actual factory form that object from the same material and recover all its original operations.

4.2 Three object domains, one universal theorem

Theorem 4.2 (One mother forms the three object domains of the original interface [7]). Fix $S_\star$. On the original interface (objects in the lowest type universe level), the following three universal statements hold simultaneously.

1. **Lawful worlds.** For every relation network N, vocabulary V, source-native authoritative root R, and lawful world state $w \in \mathsf{Law}(R)$, there is an origin record whose material rank r and material $m \in \mathcal{M}_r$ satisfy: m is formed by $S_\star$; the actual state factory forms $\mathrm{some}(V, R, w)$ from the components of m; the complete world read off is (N, V, R, w); the original material embedding has a left inverse; the registered occurrence, authoritative evolution, positive disposition, prior patches, current patch, whole-ledger write-back, next current, and advance are heterogeneously equal to those of the original object; and the root read off satisfies total reality $\mathsf{Total}(\mathcal{R}_R)$.
2. **Complete inquiries.** For every N, V and every complete inquiry state q, there is a high-rank material formed by $S_\star$ whose inquiry world read off is $(N, V, \mathrm{create}(q))$ and whose formed programme clauses equal the original clauses of q; for every query of q and every exact causal event of the original root at that visit, compilation of the clause programme is heterogeneously equal to the original query compilation; all parent material spaces are recovered by restriction.
3. **Complete macro-processes.** For every native inquiry-engine process P, there are a high-rank material m formed by $S_\star$ and a formed process P_m with

$$\mathsf{form}(m) = \mathrm{some}(P_m), \qquad \mathsf{restrict}(P_m) = P. \tag{6}$$

   State type, per-state nodes, initial state, and successor are recovered, and every query compilation at every state commutes along the same presentation; for every existing engine and activation, the complete inquiry result and the actual next engine are recovered; for every existing complete activation law, the whole law with its initial value and successor function is

recovered; for every original runtime $\mathcal{T}$ and every $n \in \mathbb{N}$,

$$\widetilde{\mathcal{T}}.\mathsf{stateAt}(n) = T_m(\mathcal{T}.\mathsf{stateAt}(n)), \quad \widetilde{\mathcal{T}}.\mathsf{tickAt}(n) \doteq \mathcal{T}.\mathsf{tickAt}(n), \quad \widetilde{\mathcal{T}}.\mathsf{tickAt}(n).\mathsf{next} = T_m(\mathcal{T}.\mathsf{tickAt}(n).\mathsf{next}). \tag{7}$$

Here T_m is the type transport induced by (6), and $\doteq$ is heterogeneous equality.

Load-bearing structure of the proof. The difficulty is coverage: a finite programme reads only finitely many observation addresses, while the target is a complete function. Given a finite input set, choose for each pair of distinct inputs an address separating them, obtaining a finite address table injective on that set; the approximation theorem of the original mother finite evaluator then approximates any target value to any precision on any finite output prefix. The basic neighbourhoods of the input-wise uniform structure consist exactly of these three data, so $\mathcal{R}_r$ is dense; the completed read-out is a uniform embedding whose image is closed and contains the dense image, hence covers all functions. Low-level addresses separate rank carriers by point observations, high-level addresses read complete low-level materials, and the same argument is used once more. The world branch pairs root material with visit material, and the original state factory recovers finite or cofinal visits by the original generation rule; the inquiry branch forms the three clause families of answer, action, and revision together with the routing that selects them; the process branch forms a complete node for every state of the original state type, together with the addresses of dependent queries, the initial value, and the successor operation. Restriction of the joint material recovers these ingredients, the actual factory forms the process from them, and restriction along the state equivalence gives (6); transporting existing activations and runtimes along (6) gives (7). The factory uses exactly the material whose formation has been proved, and downstream conclusions choose no other material. Appendix B expands the three steps and their dependencies. □

Three points keep this theorem from shrinking into "a few examples are representable". First, its quantifiers are the universal quantifiers of the original interface: every lawful world state on every network, every vocabulary, every source-native authoritative root; every complete inquiry; every complete macro-process. Second, addresses, ranks, and coverage are all constructed inside the proof; the original admission types gain no representability or coverage premise. Third, the root in the world branch is universally quantified, so any authoritative root on the original interface proposed as "another source", together with each of its lawful world states, is a read-out of $S_\star$'s completed material [7]. One fixed source already suffices to form every object on the original interface, with no object-by-object choice; nor is $S_\star$ picked from candidates: the original event structure generates a root by itself whose n-th visit reproduces, for every n, the current of the n-th visit of the original mother history [7] (Theorem 8.3).

Completed material is not runtime inventory. The completion in (5) is fixed by a universal property and is formation material; the formation theorem recovers the successor function and the runtime history at every finite depth (7), and the next actual occurrence is still generated by that successor at run time.

4.3 The same physical occurrence puts formation to work

A universal theorem that stays universal has not yet been consumed. The original physical process provides direct consumers.

Corollary 4.3 (Five direct consumers). The following five results all call the same main declaration of Theorem 4.2 directly [7]:

1. **Original world**: for any N, V, R, w, extract the material formed in the closure of the low-rank programme image, successful formation, the complete world, and recovery of the whole-ledger write-back and next current;
2. **Original inquiry**: for any N, V, q, extract the high-rank material, parent retention, the inquiry world, the clauses, and recovery of compilation at every event;
3. **Original process**: for any P, extract the complete process equation (6) and the recovery of states, nodes, initial state, successor, and compilation;
4. **Original physical seal and whole law**: for the original physical inquiry process, at the 16th state of its runtime, the transported engine answer is heterogeneously equal to the original answer, the next engine equals the transport of the original next engine, and the whole activation law with its initial value and successor function is heterogeneously equal;
5. **Original physical tick and history**: for **every** $n \in \mathbb{N}$ there is a material formed by $S_\star$ making the transported 16th tick heterogeneously equal to the original tick, with next state the transported 17th state, and recovering the n-th state, the n-th tick, and its next state item by item; the physical visit is the 10th visit of the original mother history, and same-occurrence activation holds. One material serving all n at once is given by (7) of Theorem 4.2(3).

The original formal control further checks, one by one, that the proof terms of all five results cite this main declaration directly and cite directly none of the three low-level coverage lemmas (object-wise coverage for lawful worlds and for the two kinds of source) or the four low-level recovery lemmas; that their complete dependency closure contains only the three axioms of propositional extensionality, classical choice, and quotient soundness; and that it contains the original same-occurrence activation.

Theorem 4.4 (Classical field, matter, quantum read-out, and macro next of the same living root [7] (Theorem 8.3)). From the SpinPair living root a faithful realization $\mathcal{Q}$ is generated such that: (i) for any two visits, "they form an internally determined difference" holds iff their represented visits in $\mathcal{Q}$ differ; (ii) for any other faithful realization there is a commuting presentation with which every candidate presentation agrees pointwise at every event; (iii) every event recovers all original projections, the original writer, and the whole-ledger write-back; (iv) at every event of the original physical visit, the configuration read off passes the original classical-world admission, matter recovery holds, the quantum read-out equals the answer a of the runtime tick, and for every point x and every complex-linear endomorphism A of the complete matter carrier,

$$\chi_x(A\,\psi_x) = 4k\,\langle a_x,\ R(A)\,a_x\rangle, \qquad k = \sqrt{2}, \tag{8}$$

and finally the next node of the runtime tick, after erasure, equals the next current compiled at that event.

(8) is the classical–quantum pairing equation of the physics flagship, with $R(A) = SPAE$ the response matrix obtained from the mother endomorphism through the occupied embedding E, the occupied projection P, and the Dirac upper/lower swap S; dropping S changes the pairing computed [7] (Theorem 6.2). The coefficient $4k$ is the product of the amplitude $\psi = 2Ev$ of the specified solution and the independent dual coefficient $2k$; it is not fitted. The point here is where it is **consumed**: admission of the classical field, the same matter, the quantum read-out, pairing with arbitrary actions, and the macro next hold at the same event on the same faithful realization, and (i) connects the internal determination of §2 to the physical presentation.

The physical equations themselves—the joint solution of the nine fields, the action, the quantum construction—are paid item by item by the source constructions of the physics flagship; what is consumed here are their results.

5 A Ledger Should Do More Than Hold a Grudge: Let It Pay Once

Identity and formation are in place. A reader may reasonably suspect that complete attribution is only a stamp applied after the fact. Can a ledger that only establishes "who owes what" use the material it has already generated to do the next thing itself? The answer of this section is yes. The results belong to Whole-Ledger Accounting [8] (Theorems 9.1–9.3); we write out their load-bearing argument.

5.1 Syntax, cost, and the original residual

Let S be any type, V_s an abelian group per sort, and X_s the variable types; an environment Γ gives each variable a value. Source expressions are built from variables, constants, addition, unary action of group homomorphisms, and biadditive action; evaluation is $\mathrm{ev}_\Gamma(a)$ and the source-generated increment is $D_a(\Gamma, \delta)$. The original recursive evaluation law gives

$$\mathrm{ev}_{\Gamma+\delta}(a) = \mathrm{ev}_\Gamma(a) + D_a(\Gamma, \delta). \tag{9}$$

Biadditive nodes keep the left increment, the right increment, and the cross term; (9) does not rewrite the effect of a nonlinear expression as an evaluation on δ alone.

The syntax cost $c(a) \in \mathbb{N}$ counts nodes: variable 1, constant 0; addition and biadditive action the sum of the two subexpressions plus 1, unary action the subexpression plus 1. Every lawful execution step $a \rightsquigarrow a'$ preserves evaluation and satisfies $c(a) = c(a') + 1$; a complete execution state $\sigma = (a', \tau)$ carries the ordered paid trace τ from a, so $|\tau| + c(a') = c(a)$.

Start with a nonzero bill. On the real carrier, use multiplication as the biadditive action, with $a = x \cdot x$, old environment $\Gamma(x) = 2$, and increment $\delta(x) = 2$. The old expression has executed to $p = \mathrm{const}\, 4$; in the new environment $x = 4$, the request and its evaluation are

$$r = a + (-\mathrm{id})(p), \qquad \mathrm{ev}_{\Gamma+\delta}(r) = 4^2 - 4 = 12.$$

The source increment also gives $D_a(\Gamma, \delta) = 2 \cdot 2 \cdot 2 + 2^2 = 12$, with the cross term retained. The syntax costs are $c(a) = 3$, $c(p) = 0$, and $c(r) = 5$: the completion value is 12, while executing this

request pays for 5 charged nodes. The theorems below connect these two numbers to the original stop frame, actual responsibility row, and answer.

Fix a completely admitted source, root, and temporal visit. An actual occurrence o carries an original complete ledger $L(o)$ and the next complete ledger write E_o; for any old row e, $E_o.\mathrm{dest}(e)$ gives the target row and that row's lawful evolution, with same-debt evolution never increasing budget. The **original residual material** of o is $m = (\Gamma, \delta, a, \sigma = (p, \tau), e)$, with $e \in L(o)$ the actual bearing row. The literal residual and request are

$$r_m = a + (-\mathrm{id})(p), \qquad Q_m = (\Gamma + \delta,\ r_m,\ e), \qquad c(r_m) = c(a) + c(p) + 2. \tag{10}$$

5.2 The residual is equivalent to the paid boundary at every future reading

Theorem 5.1 (The residual in the future model belongs to the next original request). Fix any original source factory $\mathcal{F}$, complete initial frame I, programme cfg, and language identity $\mathsf{cfg.LowVar} = X$. The original run produces a sequence of admission stops, indexed by $n \in \mathbb{N}$ and cofinal in the actual run ticks. For each n, let q_n be the request generated by the $(n+1)$-st complete packet from the original root current, and b_n the exact transport of the paid boundary $[a] - [p]$ of the same occurrence. Then for every finite allowed action word h and every original site $j \in \mathbb{N}$,

$$\mathrm{Obs}_j(A_h(\iota_{n+1}[q_n.\mathrm{AST}])) = \mathrm{Obs}_j(A_h(\iota_{n+1} b_n)). \tag{11}$$

ι_j is the site-preserving dependent direct-sum embedding, A_h executes the original source action letter by letter (allowed action words are the finite iterates of the original action), and Obs_j reads the complete old-value/effect pair at site j. Moreover, the actual next admission current equals exactly the canonical successor of the original birth programme, the request is generated directly from the whole material transported along that equation, and the dispositions of the old responsibility row and the born row are both read from the same whole-ledger write.

Load-bearing steps of the proof. Let Z be the submodule of integer-coefficient words that evaluate to zero in **every** environment. By the definition of evaluation,

$$[r_m] - ([a] - [p]) \in Z \tag{12}$$

for every material; what vanishes in all environments is the difference of the two expressions, not the residual itself. Let $w \in Z$: by the linear form of (9), old and new evaluations are zero, so the effect is zero and the complete inventory is $(0, 0)$. The evaluation of the literal pair lift in a pair environment is exactly this inventory, and every pair environment is recovered from its two coordinates, so the lift preserves Z; transport along the language and complete-packet equations preserves Z by equality elimination. The original action at each site is precisely the lift followed by these two transports, so it sends Z into the Z of the next site. Induct on the length of the action word: for the empty word the non-input site components are zero and the input-site inventory is $(0, 0)$; adding a letter, the original action sends $\iota_k w$ to an element of Z at ι_{k+1}. No action word and no site sees the difference (12), which is exactly the fibre equivalence of the complete future model. Finally, recognize the actual request: the registered syntax tree of the original birth is definitionally

r_{m_n}, and the next complete packet is given recursively by the language transport of the original birth packet; composing the two gives (11). □

(12) does not say that r_m vanishes in all environments. Take $a = x$, $p = 0$: in the old environment $x = 0$ the residual is 0, in the environment $x = 1$ it is 1; in both it equals the evaluation of the complete boundary. The residual that the next complete packet generates again appends a new negative paid term and costs at least 2 more, so it cannot pose as q_n either.

5.3 The same debt, actually paid

In the next two theorems, evaluation of the original residual, the first-step payment, and suffix completion hold for every original frame and programme; the readings that bring the complete old-value/effect pair back to the same current and that the consumer reads jointly are obtained on the factory $\widehat{\mathcal{F}}$ derived inside the source from $\mathcal{F}$: it only installs the complete pair reading of the same current into the computational reading of the original factory, keeps the original readings, writer, inventory callbacks, and source seed, and reads at every admission point produced by $\widehat{\mathcal{F}}$ [8] (§9.2).

Theorem 5.2 (Source-generated effect and same-debt payment). At the same current read off from the original residual material at an admission point of $\widehat{\mathcal{F}}$, let $\epsilon = D_a(\Gamma, \delta)$, $Q = c(r_m)$, and b the original birth of this stop frame. Then the registered input of b equals exactly Q_m and the evaluation of the next canonical request equals ϵ; b has executed the original birth first step, and its remaining syntax cost is $Q - 1 > 0$; the original actual action is necessarily the paid branch, generating a same-debt payment π_b whose same-debt target is unique, whose budget strictly decreases without refill, and which lands on the destination of the original row in the whole-ledger successor of b.

Load-bearing steps of the proof. The stop receipt gives the paid endpoint $p = \mathrm{const}\, v$, and soundness of the execution trace gives $v = \mathrm{ev}_\Gamma(a)$, so by (9)(10),

$$\mathrm{ev}_{\Gamma+\delta}(r_m) = \mathrm{ev}_\Gamma(a) + D_a(\Gamma, \delta) - v = \epsilon. \tag{13}$$

The material's increment is defined as the active environment minus the registered one, so $\Gamma + \delta$ is exactly the birth's active environment and the birth input is Q_m. By (10), $Q = c(a) + c(p) + 2 \geq 2$; the birth has depth 0 while the original frame reads the current after one source run, so the first step has been executed and the registered clock gives the remaining cost $Q - 1 > 0$. If the actual action were a mathematical settlement, its settlement cost would have to be 0, a contradiction; so eliminating the actual-action branch must yield a paid step, and the original payment construction consumes only this step and the equation stating that it is the actual action. Same-debt uniqueness comes from the sum type of claims: an inherited row belongs to another claim branch and cannot pose as the original debt. The strict debit comes from the actual one-step decrease of cost, no refill from monotonicity along the same row's evolution. Finally, the payment process and the enriched stock root have definitionally the same ledger root, and both finite visits recurse with the same initial state and successor, so induction on the step count gives equal visits and hence heterogeneous equality of the payment evolution with the original whole-ledger destination. □

Theorem 5.3 (Original suffix completion, answer, and canonical next). In the same notation, the remaining history of length $Q-1$ generated from the current of b has as its target exactly the completed runtime of the original source, with completion value ϵ. The original completed action is a mathematical settlement; the original whole-ledger target compiler generates for this actual completion event a debt-admission action G, which simultaneously gives an answer with its original receipt, the canonical causal answer and next current, the destinations of the old row and the born row in the same whole-ledger write, and a proof that the successor is lawful. For any result type, the original completion consumer on $\widehat{\mathcal{F}}$ directly eliminates this action, together with the heterogeneous equation "completion value equals the evaluation of the rooted material's paid state" and the "paid execution trace effect" relation.

Load-bearing steps of the proof. The original runtime satisfies $\mathcal{R}(k).\mathrm{advance}(\ell) = \mathcal{R}(k+\ell)$ and the cost law $c(\mathcal{R}(k)) = Q-k$. The birth current is $\mathcal{R}(1)$; generating $Q-1$ ticks gives $\mathcal{R}(Q)$, where the cost is 0; if the actual action were still paid, it would need a natural-number budget below 0, so it is a mathematical settlement, and by identity of the birth input its completion value equals the ϵ of (13). Executing the original birth programme at the exact temporal event of the completed root, the compiler returns the debt-admission action according to the settlement branch just obtained; its target keeps the completion event, the complete source successor, the original first step, the complete old projection, and the complete first write. The consumer first takes this action and then passes in, directly, its causal answer, receipt, completion-value relation, compilation equation, next-current equation, successor lawfulness, and complete destination evidence. No other answer, receipt, target ledger, or successor is attached from outside. □

Substitute the square example of §5.1 into this stop-frame contract: $Q = 5$, the remaining cost after the birth first step is 4, and the next actual paid action debits it to 3, landing on the original row's whole-ledger destination. The four-tick suffix generated from the birth current reaches cost 0 and delivers completion value 12, the original receipt, and the canonical next. Evaluation, cost, same-debt destination, and the next step each give a result the reader can check.

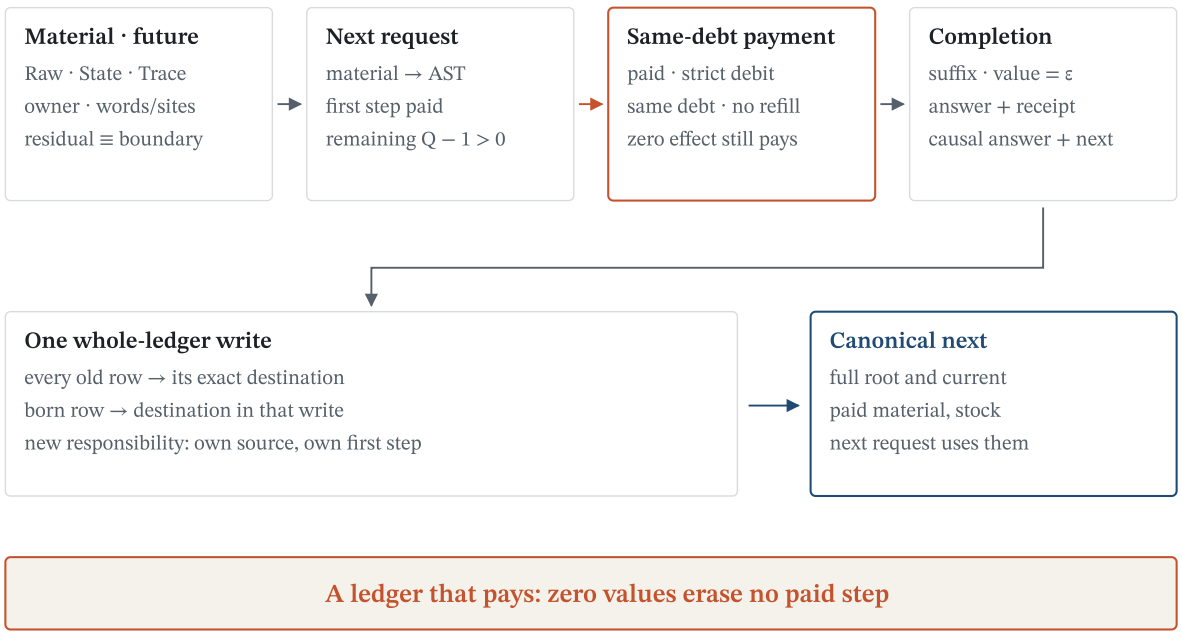


Figure 3 No free page-turns. The original material (environment, syntax tree, execution state with its complete trace, bearing row) makes the literal residual and the paid boundary equivalent at every action word and site, and generates the next request; the original first step has been executed and the remaining syntax cost is positive; the same-debt payment strictly debits without refill; suffix completion delivers the numerical completion value, an answer with its receipt, and the canonical next. Lower row: one whole-ledger write places every old row and the born row, and the canonical next keeps consuming the paid material and stock. Zero effect still pays syntax cost, and a new responsibility has its own source and first step. Arrows denote actual generation or transport along exact current equations.

5.4 Zero effect still shows a receipt; new debt does not top up old debt

Take $a = \mathrm{const}\, v$ and paid endpoint $p = \mathrm{const}\, v$. For every environment increment $D_a = 0$, yet the next request is still

$$r = \mathrm{const}\, v + (-\mathrm{id})(\mathrm{const}\, v), \qquad c(r) = 2.$$

The original first step executes the negation node, leaving cost 1; the next actual payment executes the addition node, strictly paying $1 \to 0$ and producing $\mathrm{const}\, 0$. A zero numerical effect did not delete these two actual operations, their source, or their payment lineage.

The completed disposition can also generate the next request from the same occurrence, literally $r - \mathrm{const}\, 0$ with cost 4; it pays its own first step and registers its own debt. Responsibilities and claims of the extended ledger both use sum types: the old ledger lives in the left branch, the newly born debt in the right, and the initial origin is obtained by eliminating the two branches directly, with no external "old or new" classification field. The zero-budget row of the old debt continues as the same debt by the original write; the positive budget of the new debt belongs to another source-generated row. No refill constrains the continuation of the old lineage; it does not stop the same occurrence from lawfully generating a new responsibility with its own claim, first step, and receipt. **A ledger should do more than hold a grudge: let it pay once**—with its own money.

6 Language Not Enough? Switch. Bring the Old Debts.

When expressive power runs out, a theory may revise itself. Here the reader's suspicion is sharpest: is revision just renaming the anomaly and declaring victory? This section splits "what is generated", "what is recovered", and "who disposes" into three constructions that can be checked separately.

6.1 An exact failure generates the minimal new direction

Definition 6.1 (Rooted expressibility failure). Fix an old source-native authoritative root, a temporal visit, an old theory Θ, and an obstruction o, and give a **double-emptiness** proof: the expression fibre and the lawful-realization fibre of Θ for the demand are both empty. The failure is **rooted** if it also carries the following dependent fields: Θ is exactly the law surface already owned by this source epoch of the root (theory ownership); the failure payload is the active reading of a projection installed in the root at this visit, equal to the canonical token generated by the failure, calculus, and event (occurrence projection); a source-owned obstruction calculus together with the event it generates on this obstruction, the event being equal to the calculus's canonical emission; the theory audit of that event; equality of the support at the visit with the support of the obstruction; an open responsibility row e of the old root which, after support transport, equals exactly the demand row of the event; commutation of the audit's disposition of e with the root's whole-ledger compilation of e; the causal authority of e at the visit; and the causal successor of e. A rooted failure can be constructed only from this root projection face; its constructor is not open to callers. [6]

These fields pin down "this question failed in this old theory this time": obstruction, occurrence, old row, and successor are aligned on the same root. A caller cannot attach another calculus or recognize another row after seeing the old root.

Theorem 6.2 (The minimal new direction and its initiality). Every rooted failure η generates the coordinate carrier

$$C_\eta = L_+ \sqcup \{\star_\eta\}, \qquad \beta(\mathrm{inl}\, e) = \beta(e), \quad \beta(\mathrm{inr}\, \star_\eta) = 0, \tag{14}$$

where L_+ is the complete open ledger at the old successor support. The old ledger enters C_η by the left injection, and the retraction sends the new point to the target row already carried by η, so embedding and then retracting an old row recovers it exactly; the new point has no old preimage, and its attaching direction is unique. The incidence and claim coordinates likewise each gain one new element without old preimage; the new claim holds at the new support and carries a unique semantic change from the original obstruction claim, indexed by the failure; every old expression theorem survives as the same fact on the embedded old face. For every **admissible revision**—providing constructive retractions of the old open ledger, old incidences, and old claims together with one new element without old preimage, with inherited budgets not increasing and the new element's budget zero—there is a canonical faithful morphism: all three coordinate maps are injective, preserve the old embeddings and the new element, and do not increase budgets; it is unique in the class of faithful coface morphisms, and every candidate coordinate map respecting the old embeddings and the new element agrees with it pointwise.

Proof. Disjointness of the left and right constructors rules out an old preimage of the new point; the retraction's value at the new point is the target row carried by η, not a separately chosen old row. For an admissible revision, set $F(\mathrm{inl}\, e) = j(e)$, $F(\mathrm{inr}\star_\eta) = a$, and likewise for incidences and claims; old–old equality is undone by retraction, old–new equality is excluded by "no old preimage", and new–new has a single case, so the maps are injective. Uniqueness: any candidate's value on each constructor is already fixed. □

The minimal direction prescribes not the outcome of a revision but the place every revision must pass through: **a revision may be bigger, but it cannot go around this.**

6.2 Revision must write the old rows and the new row onto the first page together

Definition 6.3 (Typed semantic revision and field grounding). For a rooted failure η, a **typed revision** Ξ provides: a non-conservative translation between the new world network and the old (old identities and old open ledgers survive by constructive retractions, new coordinates may have no old preimage); a new vocabulary and a new living root with its terminal-handoff law; a new initial support retracting onto the exact successor support of the old root; an embedding of the old target open ledger into the new initial ledger by retraction, preserving translated claims with budgets not increasing; a bidirectional correspondence between the old successor occurrence and the new initial occurrence preserving canonical emission; the canonical whole-ledger compilation of the new root's first visit as a non-terminal first write; a new row of budget zero whose claim holds at the new initial support, carries a semantic change, and has no old preimage, the incidence of the new support also having no old preimage; and **every inherited old row** together with this new row being generated rows used by the first write.

Field grounding comes in two kinds: **exact grounding** recognizes the new network, root law surface, translation, initial support, old-ledger embedding, new row, new fact, and semantic change field by field as the minimal construction of Theorem 6.2; **richer faithful grounding** does not require the new network to equal the minimal one, but requires the new law surface to be the root semantics of the new network and the first-write compilation to **actually select** a genuine split—two distinct child rows from one migrated old row, with strictly smaller child budgets, and no dormant rows in the initial ledger besides the old ledger and the new row.

A grounded typed revision is still only a candidate. To become the actual revision branch of an inquiry, it must be emitted by that fixed inquiry programme at the exact occurrence; a separately assembled candidate is not one the run has adopted.

Theorem 6.4 (Factorization of the same direction and the first write [6]). Every typed revision Ξ generates an admissible revision and its unique faithful factorization; this factorization sends the minimal direction to the new row of Ξ, and that new row is exactly the one consumed by the first write of Ξ.

Theorem 6.5 (Old responsibility and new responsibility enter the same disposition of the new root). Let Ξ be field-grounded. Then:

1. the migrated old failure row and the new row each obtain initial causal authority from **their own** initial-patch membership;

2. both pass through the **same** compiler-generated first-write successor and each obtains its target authority;
3. the new root's own causal "answer-and-next" compiler gives, at the target visit, the answer and next of the old row and of the new row; more generally, every generated initial row of the first write receives such an answer and next, and their next currents are all the same;
4. the migrated old row is exactly the demand row at this occurrence of the old root, and equals the demand row of the canonical event of the source-owned calculus; the recovery layer accepts no second recognition, and has no field for submitting a failure row after the fact.

One value carries at once the initial factorization of Theorem 6.4, the first-write attaching realization, and the two answer-and-next tokens. If Ξ is moreover actually emitted by the fixed inquiry programme at the exact occurrence (Example 6.6), this is the inquiry's actual disposition.

Proof. The old row and the new row arrive by different roads and meet at exactly one point: that first write. For each row, construct the initial authority from its row in the generated patch of the new first visit; take the first write's successor equation once and apply the same continuation to each authority, obtaining the target authorities; then call the new root's causal answer-and-next compiler. The next currents are all read from this exact revised root at the same target visit, hence equal. (4) composes the demand-row equation of Definition 6.1 with "the event equals the canonical emission". □

This separates exactly two things that are often run together:

- **Registering the anomaly**: the failure generates the minimal direction, and revision writes it, together with the whole old ledger, onto the first page of the new root. This is revision authority, not settlement authority.
- **Actual recovery**: the new root's own classifier gives, at the target visit, the old row's terminal disposition or causal successor, and also the new row's disposition. Whether the old obstruction is settled and whether the new semantic difference continues are two accounts occurring together, neither offsetting the other.

Example 6.6 (Two-tick revision [6] (Proposition 8.9)). The old world has support $\{\mathrm{false}, \mathrm{true}\}$, with one old responsibility of budget 2 at each support; the expression and realization fibres of the old theory are empty, and the obstruction sits at support true. The first tick of the canonical runtime reads the revision branch: the minimal extension gives the new support $\star_\eta$ and the ledger "old row 2, new row 0"; the revision realization takes natural numbers as currents with native write $n \mapsto n + 1$, exactly grounded. The first write $0 \to 1$ consumes both rows, carrying $2 \to 2$ and $0 \to 0$ respectively; compilation at the target gives the same next current 2. The second tick answers the singleton value directly at visit 2 of the same new root and advances to 3. The old debt did not vanish at the seam, and the new row is not left hanging beside the semantic-change certificate.

Even revision has to pay the old debts.

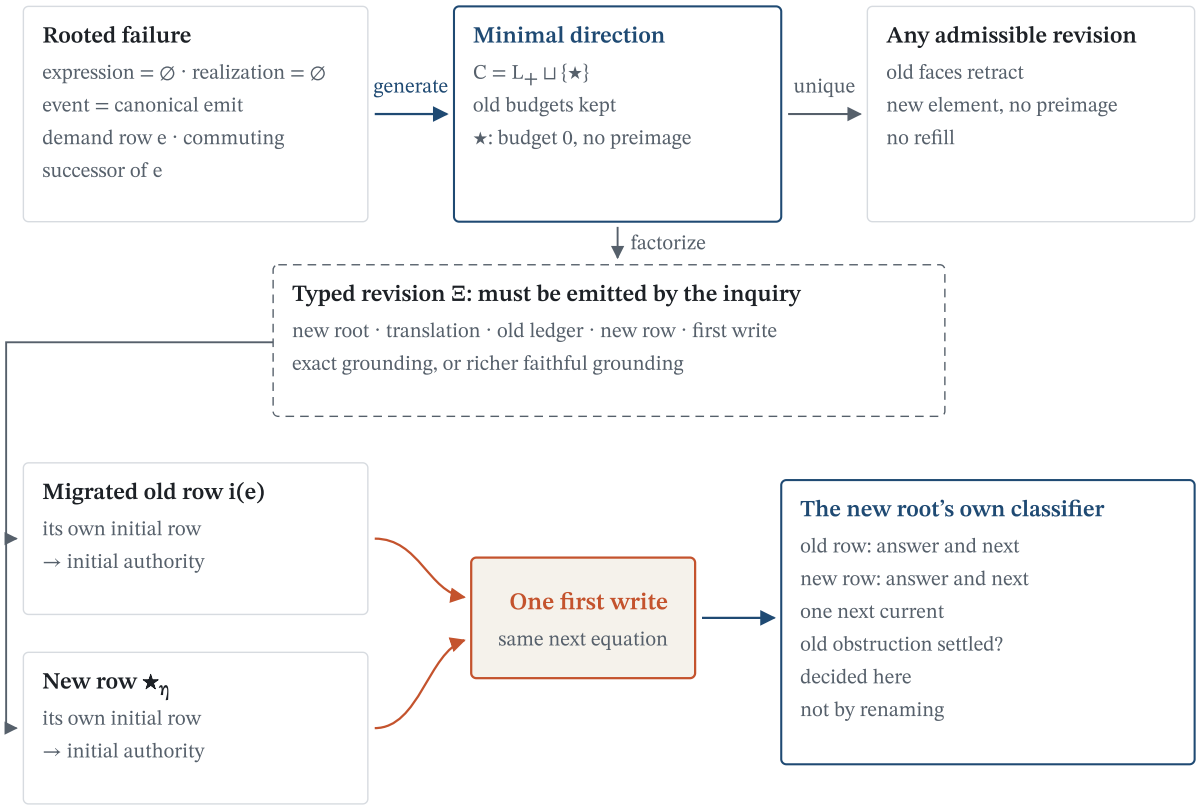


Figure 4 Amend the constitution, not the debt. Top: a rooted failure (doubly empty fibres, emitted event, demand row, commuting disposition, and successor) generates the minimal direction $L_+ \sqcup \{\star\}$; every admissible revision receives it through a unique faithful morphism. Middle: the typed revision is actually emitted by the fixed inquiry programme and field-grounded (the dashed box marks programmes and proofs the source must actually provide). Bottom: the migrated old row and the new row each obtain authority from their own initial rows, enter the target through one first write, and receive separate answers and nexts from the new root's own classifier, with the same next current. Arrows are labelled generation, the unique faithful morphism, factorization, and the one first write; there is no arrow saying "the anomaly was absorbed".

7 Long Computations Do Not Get to Drop Their Baggage Either

The same contract also connects to a finite-window analytic programme [8] (Theorems 14.1–14.2): the four traces of the first execution—forward stock, inverse residual, second mean, and cochain—enter the next correction packet; the residual is read from the original write and becomes the next computation's input. The new execution delivers its own payment, complete stock, and canonical next, and every allowed future reading continues to identify the same request and paid boundary. The construction holds universally over the stated original input type, which includes a nontrivial zero-observation receipt carrying $\operatorname{Re} z > 1/2$. Appendix E retains the analytic objects, all input conditions, and the full wiring theorem; the general payment and revision conclusions are borne by their respective contracts in §§5–6.

8 If You Want to Break It, Break This Equality

A principle that never concedes is worthless; so is a principle that only chants "falsifiable". This section lists the places where one can strike: each is a specific equation or type, and a hit refutes exactly that claim while the rest stands.

8.1 Three layers of checking

Layer one: structural identity, same-occurrence recovery, and commuting compilation. Targets: the occurrence-to-state step of Lemma 2.2, the injective complete current of Theorem 2.8, (6)(7) and the field-wise heterogeneous equalities of Theorem 4.2, and (8) of Theorem 4.4. A counterexample must be a concrete object of the **declared input type** that makes the equation fail. These equalities all have kernel-checked proofs, so the realistic point of attack is usually translation: does the natural-language text match the formal declaration exactly? If it does not, what gets corrected is that sentence and its consumers, while the formal result stands; if it does and the object you care about fails admission, name the violated admission field—the theorem says nothing about unadmitted objects, though Theorem 4.2 already quantifies universally over every authoritative root on the original interface.

Layer two: actual consumption of generation, payment, and revision. Targets: (11)(13), the strict debit, completion value equal to effect, the consumer of the completion action, and the first write shared by the two rows in Theorem 6.5. What must be checked here is "the same": the payment lands on the original row's destination and not on another ledger, the answer belongs to this completion event and is not attached from outside, the old row and the new row read the same first write. A case satisfying the original inputs in which the consumer reads a different object would hit this layer.

Layer three: domain interpretation and empirical contracts. Maps from formal objects to physical fields, molecules, and observation records, and the empirical predictions they give, are each borne by the interpretation maps and experimental contracts of the paper they belong to. A physical reading that disagrees with experiment corrects that map or that prediction; it does not revoke the payment theorems of layer two, nor is it paid for by the negative conclusion of layer one.

8.2 No renaming a failure

Whichever layer is hit, correct that actual claim and keep the original facts and responsibilities. A refuted equality cannot be renamed a "new expressibility failure" and so enter the revision path: revision requires all the fields of Definition 6.1, and a wrong equality has no emitted obstruction event, demand row, or commuting disposition. Conversely, an unclassified case cannot default to safe (§3.2).

Appendix D lists each claim's original declaration, public target, and check command: kernel checking, metadata trust, and complete type review answer "is the proof closed", and domain contracts answer "do the readings match".

8.3 The prehistory of this bit

Sufficient reason. In §§31–38 of the *Monadology*, Leibniz takes contradiction and sufficient reason as the two great principles of reasoning, requires that no fact hold and no statement be true without a sufficient reason why it is so and not otherwise, and distinguishes truths of reasoning from truths of fact [1]. This paper shares the question "why so and not otherwise". What it gives is not an ultimate reason for the whole contingent series, but, in an explicit source structure, a positive structural identity for each actual difference; structural attribution can be a simultaneous relation, a global constraint, or a registered occurrence, and need not be an earlier temporal cause.

It from bit. Wheeler holds that every physical "it" ultimately derives its significance from apparatus-elicited yes-or-no answers [2]. The link between binary question-answering and physical existence has a clear prehistory. The bit of this paper asks something else: whether an unaccounted difference can acquire standing. Its positive content is paid by occurrence, formation, and successor; observers and apparatus are in the same source.

The bit as a unit. Shannon measures choice in binary digits and the uncertainty of a random variable by entropy; a certain outcome has zero entropy [3]. A binary interface, storage capacity, and the entropy of a random variable are different concepts; the bit of this paper is of the first kind.

Proof-carried standing. In Necula and Lee's proof-carrying code, the receiver regenerates the safety predicate from the code actually received and checks the attached proof to decide whether to execute [4]. Checkable proof bearing execution standing is a concrete precursor of this paper. The contract here further binds source, occurrence, the whole ledger, responsibility lineage, and the actual successor together.

The title salutes *Attention Is All You Need* by Vaswani et al. [5].

9 Ask Once a Day; Tomorrow Still Has to Settle Up

Back to the opening bet. The two Booleans are still there, and as raw data they still differ; only now the question "are they two worlds?" has an executable answer: look at the registered occurrence.

This bit is weak. It does not foretell tomorrow, rewrite yesterday, or choose anyone's answer. It only requires every actual difference to produce its provenance. In this paper that requirement has paid out four times: unequal registered occurrences give structural identity; one fixed mother forms every world, inquiry, and process on the original interface and carries that identity back field by field; the original ledger generates a request from the complete future residual, pays the same debt with its own syntax, and delivers an answer, a receipt, and the next ledger; when language runs out, failure generates the minimal new direction, and old debt and new responsibility each receive their disposition in the same first write. For its listed inputs, a long-computation construction carries its old execution traces into the next correction; an original integer class hides from every finite complex reading, and the original consumer reads it all the same.

What remains open on the books are responsibilities that each have a name: domain interpretation maps, empirical contracts, and the tomorrows not yet generated by an actual run. None of them has been declared paid, and none has been hidden.

The bet goes on. Once a day I ask one question: where did you come from?

One bit is weak. The world that honours it is not allowed to slack.

Appendix A How This 0 Was Earned

This appendix gives complete argument outlines for the results of §§2–3, in the notation of the main text. The original formal declarations are in the fixed research source Y (see Appendix D).

A.1 The two types of temporal reachability

Given a root R, **post-cofinal reachability** is an inductive type: constructor $\mathsf{cofinal}(u)$ takes a source-generated cofinal visit u and reaches the current of u; constructor $\mathsf{step}(h, eq)$ takes a post-cofinal history h reaching c and an equation $\mathsf{next}_R(c) = \mathrm{some}\, c'$ and reaches c'. **Temporal reachability** has only two constructors: $\mathsf{finite}(h)$ with h a finite reachability history of the root, and $\mathsf{postCofinal}(h)$. The successor advances on each branch separately, with no third spelling. Since a post-cofinal history can only begin with $\mathsf{cofinal}$, it cannot be constructed from a finite visit; every later successor keeps this boundary.

The finite embedding $v \mapsto \mathrm{finite}(v)$ is injective: if $\mathrm{finite}(v) = \mathrm{finite}(v')$, destructuring the records on both sides field by field gives equal currents and equal finite histories. In Proposition 2.3(1) the history of v_1 is the initial constructor and that of v_2 the one-step constructor, and case analysis on the equation gives a contradiction.

A.2 Where each field of the root semantics comes from

In the semantics (4) of Theorem 2.6: both witnesses of actuality are proofs that "the temporal evolution generated by the root at this state equals the canonical image of the temporal-visit authority compiler"; both witnesses of stable reference say that there is an exact temporal event equal to that side's registered occurrence, taking the registered occurrence itself; the real difference is the inequality of states. The location function of total reality returns only the right branch for every actual difference, its proof being the contrapositive of Lemma 2.2. The exclusion of external supplements and magic bits is the two-branch argument of Theorem 2.5. The whole construction accepts as input no NoMagic, no external-supplement exclusion, no difference classifier, no occurrence inequality, and no second causal-closure token.

A.3 Complete living processes and current reflection

The generated-successor relation of Definition 2.7 is a single proposition, not two independent receipts: the erased target equals the source-generated next current, **and** the living root is heterogeneously equal on non-terminal branches, with no further condition on the terminal branch. This design keeps the terminal branch's second component from being used apart from the exact handoff target.

Proof of Theorem 3.1(1): from $\iota s = \iota s'$ and injectivity, $s = s'$; after case analysis on this equation, the canonical answer-and-next on both sides is the same term. The carrier of canonical answer-and-next has a single constructor, and every element of that type equals the canonical image; so "attaching another answer from outside" has no place in the type. The law-surface equation of (3) maps the successor equation by $\rho \mapsto \rho.\mathrm{lawSurface}$ and then uses the lemma that the root's generated next current preserves the law surface.

A.4 Observation kernels and the original integer consumer

The scope observer of Theorem 3.3 filters each live responsibility by the visibility of its scope into a list of lineages; the complete current-responsibility observer reads each slot as a live responsibility or empty. After admitting a commitment of invisible scope, the former is unchanged while the latter grows by one.

In the original integer instance, $G = \{(g_n) \in \prod_n C_n : \rho_n(g_{n+1}) = g_n\}$ is the globally compatible integer cokernel, $F(g) = (1 \otimes \pi_n(g))_n$, and $J(g) = \pi_1(g)$. Whole-Ledger Accounting proves: compatibility of the endpoint classes gives $z \in G$; the original unit arithmetic root gives $J(z) = z_1 \neq 0$; stagewise tensor cancellation gives $1 \otimes z_n = 0$, without exchanging limits and tensors [8] (Theorems 13.1–13.3, Lemma 13.2). If $J = h \circ F$, then $J(z) = h(F(z)) = h(F(0)) = J(0) = 0$, a contradiction. What is excluded is every function, not only linear decoders.

Appendix B Formation Does Not Smuggle the Target into the Input

B.1 From finite observations to the complete read-out

Let address observations separate the points of X_r. Given a finite input set $F \subset X_r$, choose for each pair of distinct points an address separating them, obtaining a finite address table on which the observation vector is injective on F; padding to a stream keeps injectivity. For any target $f : X_r \to \mathbb{S}$, one may prescribe target values on the finite observation image and extend to the other input streams. The approximation theorem of the original mother finite evaluator gives: for every $b \in \mathbb{N}$ and $\varepsilon > 0$, there are an actual mother visit and a finite address table with

$$|L_r(p, x)(j) - f(x)(j)| < \varepsilon \qquad (x \in F,\ 0 \leq j < b). \tag{B.1}$$

In the formal proof the visit is the $(10 + c)$-th visit of the original mother history, with c the index of the finite programme. The basic neighbourhoods of the input-wise uniform structure consist exactly of F, b, and ε, so $\mathcal{R}_r$ is dense. The completed extension $\mathsf{read}_r : \mathcal{M}_r \to (X_r \to \mathbb{S})$ of the inclusion of the original programme image is a uniform embedding whose image is closed and contains the dense image, so it is bijective [7] (Appendix C.9).

On top of this, the formation predicate of Definition 4.1 is a theorem: both the low-rank and the high-rank finite-programme images equal the raw image, which is dense in the completion, so every material lies in the closure of the finite-programme image. The formation predicate is therefore neither an address premise nor a representability premise.

B.2 All materials enter one witness

For a macro-process P, put all states and all dependent queries into the disjoint union

$$\mathcal{U}_P = \mathrm{State}(P) \sqcup \coprod_{s \in \mathrm{State}(P)} \mathrm{Query}(P.\mathsf{stateAt}(s)). \tag{B.2}$$

The cumulative material construction chooses a sufficient rank from within $\mathcal{U}_P$ and generates faithful addresses. It forms a complete query node for each original state separately, together with

the state domain and operation materials. Answered, not-yet-visited, and equally-read but distinct states each keep their own position in (B.2). The inquiry branch encodes the original query type and complete branch indices in the same way, forming the three families of answer, action, and revision together with routing, so all five complete branches of original query compilation are covered. The joint construction embeds each family of materials into a common material rank and merges them by a fixed stream pairing; the two read-outs of the pairing recover the two complete streams, and so restriction to parents is a left inverse of the combining map on the whole material space.

B.3 The factory reconnects to the original downstream conclusions

The macro-process factory reads the state domain, complete node family, event addresses, and operation graph from the joint material; by the recovery equations of the previous step, the original registered factory successfully forms a state-relabelled version P_m of the original process, and restriction along the generated state equivalence gives $\mathsf{restrict}(P_m) = P$. This complete record equation is the common basis of the downstream conclusions: equality of per-state nodes makes all queries compile equally after transport; equality of complete processes recovers existing engines, existing activations, and whole activation laws after transport; every step of every original runtime, with its next state, is recovered, giving (7) at every depth. The total-reality field of the world branch is generated directly from the root read off from the origin record, an instance of Theorem 2.6 at that root.

B.4 The five consumers and their control

Each of the five results of Corollary 4.3 cites the main declaration of Theorem 4.2 directly in its proof term and extracts its fields item by item: the original world takes formation, successful formation, the complete world, the whole ledger, and the next current; the original inquiry takes formation, parents, the inquiry world, clauses, and per-event compilation; the original process takes the complete process equation with its four structural recoveries and compilation; the original physical seal evaluates inquiry recovery and law recovery at the 16th runtime state; the original physical tick evaluates tick recovery at the 16th state and history recovery at every depth, and finally reads off the 10th mother visit and same-occurrence activation. The original control checks that each result's proof term cites the main declaration directly and cites directly none of the three low-level coverage lemmas or the four low-level recovery lemmas; that the dependency closure has no unsafe or partial declarations, its axioms being only propositional extensionality, classical choice, and quotient soundness; and that it contains the original same-occurrence activation.

Appendix C The Original Debt Was Actually Paid

C.1 The all-environment zero submodule is preserved by future actions

Write $\mathrm{Fm}(V, X; s)$ for the free module of integer-coefficient complete expression words and ev_Θ for linear evaluation. $Z = \{w : \forall \Theta,\ \mathrm{ev}_\Theta(w) = 0\}$ is closed under addition and integer multiples, hence a submodule. The original update-evaluation square gives $\mathrm{ev}_{\Gamma+\delta} = \mathrm{ev}_\Gamma + \mathrm{eff}_{\Gamma,\delta}$, so for $w \in Z$ we have $\mathrm{eff}_{\Gamma,\delta}(w) = 0$ and the complete inventory $(\mathrm{ev}_\Gamma w, \mathrm{eff}_{\Gamma,\delta} w) = (0, 0)$. The literal pair lift sends a constant v to $(v, 0)$, preserves variables and addition, sends a unary homomorphism f to $(u, du) \mapsto (fu, f\,du)$, and sends a biadditive action to

$$((u, du), (v, dv)) \mapsto \big(B(u, v),\ B(u, dv) + B(du, v) + B(du, dv)\big),$$

so its evaluation in $\mathrm{pairEnv}(\Gamma, \delta)$ is exactly $(\mathrm{ev}_\Gamma, D(\Gamma, \delta))$. Every pair environment Θ equals the pairEnv formed from its two coordinates, so the lift sends Z into the Z of the pair language. Each site action is the composite "lift, language transport, complete-packet transport", and all three preserve Z. Induction on action words gives (11); the residual relation (12) is verified directly by unfolding evaluation and abelian group operations.

C.2 Why the payment cannot be a settlement

The registered clock of the birth, obtained by induction over stepwise source runs, gives $c(\mathrm{state}(b)) = c(b.\mathrm{AST}) - (b.\mathrm{depth} + 1) = Q - 1$. By (10), $Q \geq 2$. If the actual action of the birth were a mathematical settlement, its settlement condition would require remaining cost 0, contradicting $Q - 1 > 0$; so eliminating the actual-action branch yields a paid step. The debt current in the payment takes the original mathematical row, and the same-debt lineage is constructed by induction along the original complete-ledger destinations; every target row with this claim belongs to the original mathematical debt branch, while inherited rows belong to another claim branch, so the same-debt target is unique. The identification of the payment evolution with the whole-ledger destination first eliminates the support equation and then the identity of ledger evolutions; it does not force two complete roots with different projection cofaces to be equal.

C.3 Data dependencies of revision

The dependency chain of Theorem 6.5 is

initial generated row $\longmapsto$ initial-row authority $\longmapsto$ first-write target authority $\longmapsto$ answer and next of the new root.

The second arrow uses the non-terminal proof of the first write and the whole-ledger row evolution; the last arrow calls the new root's compiler and accepts no externally supplied settlement or next current. For every initial row generated by the first write, the next current of the resulting answer-and-next is definitionally that of the old row. The original regression control also checks that the public interface of the recovery layer has no "after-the-fact failure row" field and that the failure row of the old root equals the demand row of the canonical event of the source-owned calculus.

Exact and richer grounding differ in whether the semantic fields equal the minimal construction item by item; both consume the same generated occurrence. Richer grounding requires the split actually selected by the compiler: two distinct target entries from the migrated row of the one original causal responsibility, sharing one complete split cover, with each child budget strictly below its parent; if the exact target ledger has at most one entry, or the compiler's split-cover type is empty, this two-child action type is empty.

Appendix D Where the Reader Hands Over to the Machine

Claim	Text	Original formal declaration (fixed research source)	Public check
OB1 meaning of the bit	§1.1	Y: definition of `MagicBitAt`; Shannon's original text	Definition and source reading
OB2 Boolean ghost	§2.2, A.1	Y: the ghost construction and `no_ghost_argument_in_world_state_producer` in `LivingLawZeroLawRootAdmissionRegression`	One-bit control package `HOmework.Checks.OneBit`
OB3 total reality	§§2.3–2.5, A.2	Y: `TotalRealityAt.noMagicBit`, `registeredOccurrence_ne_of_ne`, both `isTotal`, `noMagicBit`; regressions `root_rejects_magic`, `living_process_rejects_magic`	Whole-ledger package WA1
OB4 rigidity and history	§3.1, A.3	Y: `canonicalAnswerAndNext_heq_of_current_eq`, `successor_difference_reflects_current`, `successor_lawSurface_eq`, `generateHistory`	Process core C3 and whole-ledger package
OB5 observation counterexamples	§3.2, A.4	Y: `ResponsibilityObserverKernel` and lifecycle regression; original integer consumer separation	Whole-ledger package WA6, WA11
OB6 three-domain formation	§§4.1–4.2, B.1–B.3	F: `root_admission_fixed_mother_complete_realization`, `low_formation`, `high_formation`	First-release map phys.P22
OB7 physical consumption	§4.3, B.4	F: five direct consumers; H: `source_realization_consumed`	First-release map phys.P6, phys.P22
OB8 payment and answer	§5, C.1–C.2	MV4: `actual_root_future`, `actual_root_eliminate` → `consume_rooted` → `consume_rooted_completed_answer`	Whole-ledger package WA38, WA39
OB9 revision	§6, C.3	Y: `root_obstruction_generates_minimal_coface`, `minimal_coface_initial`, `minimal_coface_factorization_unique`, `generatedOldTargetAnswerAndNext`, `generatedNewTargetAnswerAndNext`; regression `generated_initial_entry_next_is_recovery_next`	Whole-ledger package WA9
OB10 long computation	§7, Appendix E	MV4: `actual_finite_window_whole_next_future`, `consume_residual_source_receipt`, `actual_residual_whole_next_future`	Whole-ledger package WA40
OB11 failure location	§8	Inputs, outputs, and commuting relations of OB2–OB10	Argument of this paper

The fixed research sources are Homework: Y = `b81a443863f4d7135ebaca92c5bb7d208ccc18ca`, H = `e60a86058abcb74b7f2225f2383c182be8343785`, F = `cde4b910929d9bc57554aa56f6ade116bc3b8675`, MV4 = `ba7e274beb1de38900a6fc2c8c8dfe6cfad9d9aa`. Each result keeps its own source epoch.

Every item is located through the one-bit map of H0mework `6bff9d4046b3f89004678244482b1c403837150c`, and `make check-one-bit-entry` checks their source digests.

Appendix E The Full Wiring Contract for a Long Computation

This appendix develops the wiring contract summarized in §7. The long-computation construction of Whole-Ledger Accounting [8] (Theorems 14.1–14.2) carries four execution traces of a finite window computation into the next correction. The analytic objects of the window—the Hilbert carrier, defect and Fourier operators, the Pa qualification projection, finite cutoffs, and the two mean identities—are supplied by the common-source constructions of the Riesz source calculus, whose analytic proofs belong to their own results. We state the original inputs first, then give the full contract for programmes, costs, inventories, and successors.

E.1 Objects and input domain

Let $\mathcal{H}$ be the original complex Hilbert carrier, D the original defect operator, and $\mathcal{F}$ the original Fourier operator, with the four-component carrier and embedding

$$\mathcal{B} = (\mathcal{H} \times \mathcal{H}) \times (\mathcal{H} \times \mathcal{H}), \qquad \iota(f) = ((f, Df), (\mathcal{F}f, \mathcal{F}Df)), \qquad \pi\iota = \mathrm{id}. \tag{15}$$

The original Pa source is $\nu = M\mathcal{E}\pi(S)$, with $\mathcal{E}$ the original qualification projection and $S \in \mathcal{B}$ the actual input of this source chain. The finite source window is

$$W_a(f) = \sum_{m \in L_a} \frac{1}{m} A_{a,m} f - \frac{1}{2}\, \ell(Jf)\, c_a, \tag{16}$$

with L_a the finite list of original centered Möbius cutoffs, $A_{a,m}$ the scale pull-backs, ℓ the source window integral, J the Tate reciprocal action, and c_a the constant interval field. The original inputs of this source chain are a zero-observation receipt z on the original analytic occurrence (carrying $\zeta_{\mathrm{src}}(z) = 0$, nontriviality $z \neq -2(n+1)$, and the right-half condition $\mathrm{Re}\, z > 1/2$), together with arbitrary depth, ordinal, and real parameters h, H, N, a. The construction holds universally over this input type; if the Riemann hypothesis holds, the type is uninhabited and Theorem E.1 below holds vacuously. The sum, the reciprocal weights, and the Tate pole term all stay in the programme as syntax nodes: when $|L_a| = k$, the window programme costs $4k + 3$.

E.2 Two occurrences, one ledger

Theorem E.1 (Four prior traces enter correction, payment, and every future [8]). For every original input as above:

1. the difference Δ_F of the first actual write equals $\iota Z_a W_a(\nu)$; the four original execution inventories—forward stock, inverse residual, second mean, and cochain—are installed before birth; reading from this write,

$$\eta = \nu - MP\pi(\Delta_F), \qquad r = \iota(\eta), \tag{17}$$

η equals the complete residual generated by the first inverse programme. The window, the two original means, and every execution entry enter this birth packet, whose completion receipt gives its own writer, same-debt payment, and exact whole-ledger destination.

2. The next source executes the literal correction $\kappa = \operatorname{const} s + (-\mathrm{id})(\operatorname{const} r)$ (two charged nodes, cost 2), substitutes it into the original complete window programme, and generates a correction packet with its completion receipt, inventory, and whole-ledger write, with

$$\Delta_C = \iota Z_a W_a(\nu - \eta), \qquad R_a \pi(\Delta_C) - W_a(\nu) = -W_a(\eta). \tag{18}$$

3. The four traces of the first occurrence enter the new packet by the original inventory migration; the correction frame also actually generates its own four classes of window execution traces, together with a new stock trace and correction trace. The old-write proof, the new-write proof, the new whole-ledger payment, the original environment/syntax-tree identity, and the actual execution traces are eliminated in nested fashion in one consumer.
4. The birth root and current are recognized as the root and current of the complete packet; continuation starts from the packet's actual source seed, which no auxiliary seed can replace; for every allowed finite future word, every future admission ordinal, and every site, the readings of the original request and its paid boundary are equal.

What this construction shows is the same contract written onto a large programme: the residual is not merely kept, it becomes the input of the next piece of actual work; the window value is not a construction input but is read off from the received execution traces; and no inventory cost is passed off as the budget of the original debt. **Long computations do not get to drop their baggage either**—they have to burn it as the next run's fuel.

Code and Data Availability

All formal material of this paper is located through H0mework commit `6bff9d4046b3f89004678244482b1c403837150c`, with material commit `c40ef754ccbbea96f8c1417f12348e840da5eb82`. The one-bit map connects OB1–OB11 item by item to the original declarations, exact historical environments, and evidence; readiness records its acceptance. From the repository root, `make check-one-bit-entry` checks the complete source and import digests, original named audit roots, and selected resources of the five accepted original packages (whole-ledger Y/MV4, first-release core H, physics H/F), and restores the selected sources exactly from public modules; it runs no scientific computation.

The original formal control for the Boolean ghost of §2.2 is public as `H0mework.Checks.OneBit.ZeroLawRootAdmissionRegression`; its original Y source has SHA-256 `59132da0687ab67b235ac4fdffb9fe585ed02c104787571ee5f1d0928f2aaf06`. After `make bootstrap`, `make check-one-bit-full` runs a minimal-closure build, complete metadata trust, and complete type check on it and on the admission declarations it consumes directly: 17 control declarations and 6 admission declarations pass, with an empty axiom dependency and no unsafe or partial declarations. Two of them use `fail_if_success` to check that the illegal terms fail to type-check, and are recorded separately from the equality/inequality controls. Full commands and original-path restoration are in the reproduction guide.

The remaining items reuse the valid acceptance of their original packages: OB3, OB5, and OB8–OB10 belong to the whole-ledger package (item map; complete rebuild with `make check-whole-ledger-full`), and OB6–OB7 belong to the first-release physics packages (phys.P6 and phys.P22 of the first-release map; complete rebuild with `make check-first-release-full`). The toolchain is fixed at Lean 4.33.0 / Mathlib v4.33.0 by `Lean/lean-toolchain` and `Lean/lake-manifest.json`.

Shared process results cite the process core, second edition, DOI 10.5281/zenodo.23297175; physical formation cites the physics flagship, second edition, DOI 10.5281/zenodo.23297179. Figures come with editable vector sources and their generator. Original code is licensed Apache-2.0; the paper and original figures are licensed CC BY 4.0.

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